

Weak-proxy

Token overlap, Edit similarity, AST node similarity, API overlap, API mismatch

Lexical

Length ratio

Syntactic

Parse validity

Normalized-structure

AST depth similarity, AST shape similarity, LN function count, LN class count, LN max AST depth

Normalized-control

Control flow, Nesting depth, LN CFG node count, LN CFG edge count, LN cyclomatic complexity

Normalized-operator

Condition/operator patterns, LN operator families

Semantic-static

Identifier overlap, Identifier role consistency, Data-flow approximation, Type consistency

Normalized-program

LN syntax validity

Normalized-call

LN call count

Normalized-identifier

LN identifier count, LN identifier entropy, LN identifier role count, LN identifier-role distribution

Normalized-dataflow

LN def-use count, LN read-write ratio, LN assignment density

Dynamic

Example execution, Full execution